

ENGLISH FOR WORK / SEPTEMBER 2026

Semiconductor English

Vocabulary & phrasebook

Keep precise meanings and useful expressions close. Retrieve the words, then use them in a complete message.

PRACTICE THAT TRANSFERS TO WORK

Read a realistic case. Find the right language. Speak, write, get feedback, and try again.

Eight lessons | Intermediate to advanced | Independent or classroom study

For semiconductor process engineers, yield engineers, product engineers, equipment engineers, fab supervisors, quality teams, test and packaging teams, foundry coordinators, supply planners, applications engineers, and technical program managers.

These cases practice communication. Actual equipment, construction, and safety work follows approved procedures and authorized specialist decisions.

All scenarios, figures, and model responses are fictional teaching material. Use invented details in practice.

[Open this course on English Ladder](#)

Make vocabulary usable

Knowing a definition is a start. To use a term well, explain it in plain English, connect it to a case, and choose a natural sentence around it.

- Read five terms, cover their definitions, and recall the meanings.
- Sort a pair you might confuse and explain the difference.
- Use one term in a sentence about a fictional case.
- Ask a partner to explain the sentence without repeating the specialist word.
- Return to the same terms on another day and test recall again.

Abbreviations need context

Expand an abbreviation on first use when the listener needs it. Some abbreviations have different meanings across professions: API can refer to an application programming interface or an active pharmaceutical ingredient. The course context determines the meaning.

Definitions have limits

These are concise language-learning explanations, not complete legal, clinical, or technical specifications. Use the applicable authoritative definition for regulated or operational decisions.

Field vocabulary A-Z

binning

Sorting tested semiconductor units into performance, power, speed, or quality categories.

burn-in

Stress testing used to screen for early-life failures before product release or shipment.

capacity allocation

Decision process for assigning limited foundry, tool, test, or assembly capacity across products or customers.

cleanroom

Controlled manufacturing environment designed to limit particles, humidity, electrostatic risk, and contamination.

CMP

Chemical mechanical planarization; a process that smooths wafer surfaces for later manufacturing steps.

contamination control

Practices used to prevent particles, residues, metals, organics, moisture, or handling errors from affecting wafers or devices.

critical dimension

A measured feature size on a wafer that must stay within specification for device performance and yield.

defect density

Number or rate of defects on a wafer, die, layer, lot, or process area.

deposition

Process of adding material layers to a wafer by physical, chemical, epitaxial, or atomic-layer methods.

etch

Process that removes selected material from a wafer using wet chemistry or plasma-based methods.

excursion

Manufacturing event or trend outside expected control limits, specifications, or normal process behavior.

fab

Semiconductor fabrication facility where wafers are processed through manufacturing steps.

foundry

Semiconductor manufacturer that fabricates chips for external customers or design companies.

lithography

Patterning process that transfers circuit features to a wafer using light, masks, and photoresist.

metrology

Measurement discipline used to verify process, dimension, film, defect, and device characteristics.

node

Technology generation or process family, often associated with feature size, performance, density, and design rules.

package

Protective and electrical interface that connects a semiconductor die to a board or system.

particle

Small contaminant that can create defects, yield loss, reliability risk, or process instability.

PDK

Process design kit; foundry-provided design rules, models, and files used to design chips for a process.

photoresist

Light-sensitive material used in lithography to define patterns on a wafer.

preventive maintenance

Planned equipment service performed to reduce unplanned downtime, drift, contamination, safety risk, or tool instability.

process flow

Ordered sequence of semiconductor manufacturing steps, layers, inspections, holds, and decision points.

process window

Range of process conditions under which results meet specification with acceptable margin.

qualification

Evidence-based approval that a process, product, tool, package, supplier, or change meets defined requirements.

recipe

Controlled equipment parameters used to run a process step on a wafer, lot, or tool.

reticle

Photomask used in lithography to project circuit patterns onto a wafer.

SPC

Statistical process control; use of control charts and limits to monitor process stability.

tape-out

Final release of a chip design to the foundry for mask generation and fabrication.

tool matching

Effort to make similar manufacturing tools produce equivalent results within defined limits.

tool uptime

Percentage of time equipment is available and qualified for production use.

wafer

Thin semiconductor substrate, usually silicon, on which integrated circuits are fabricated.

yield

Share of wafers, die, units, or lots that meet requirements after manufacturing, test, or qualification.

Specialist terminology

Additional semiconductor terminology grouped by technical area.

Fab logistics

FOUP

Front opening unified pod used to transport and protect wafers in automated fab environments.

lot traveler

Record or system route showing where the lot goes, what has been done, and what instructions apply.

hold code

System code that explains why material is blocked from movement or shipment.

WIP

Work in process; material currently moving through manufacturing, test, assembly, or disposition.

queue time

Time a lot waits between process steps; excessive queue time can affect quality or cycle time.

lot genealogy

Traceable history of tools, materials, routes, rework, holds, and related lots.

Process control

control wafer

Wafer used to monitor process or tool behavior without risking production material.

split lot

Lot divided or compared across conditions to test a process variable or disposition path.

golden tool

Reference tool whose performance is treated as the comparison point for matching.

chamber matching

Effort to align performance across chambers in the same or similar equipment.

inline monitor

Measurement taken during manufacturing to catch drift before final test or yield loss.

PCM

Process control monitor structures or measurements used to evaluate process health.

Lithography**overlay**

Alignment accuracy between patterned layers on the wafer.

dose

Exposure energy applied during lithography.

focus

Exposure focus condition that affects printed feature quality.

OPC

Optical proximity correction used to adjust mask patterns for printing behavior.

CD-SEM

Critical-dimension scanning electron microscope used to measure small patterned features.

scatterometry

Optical metrology method used to infer feature profiles, dimensions, or film characteristics.

Films and etch**film thickness**

Measured thickness of a deposited or grown layer.

sheet resistance

Electrical resistance measurement used to monitor thin films or implants.

selectivity

How much faster one material is removed than another during etch or CMP.

uniformity

Degree to which a process result is consistent across wafer, lot, tool, or chamber.

end-point detection

Method for determining when an etch or process step has reached the target condition.

slurry

Abrasive chemical mixture used in CMP.

post-CMP clean

Cleaning step used to remove residue, particles, or contamination after CMP.

Test and sort**wafer sort**

Electrical testing of die while still on the wafer.

die sort

Classification of die after wafer-level testing.

probe card

Test interface that contacts wafer pads during wafer sort.

parametric test

Electrical measurement of device or process parameters, often used to monitor margins.

guardband

Extra margin used to reduce risk when specification, process, or use conditions vary.

Reliability**infant mortality**

Early-life failures that appear soon after device use or stress.

FIT rate

Failures in time; reliability measure often expressed as failures per billion device hours.

electromigration

Reliability failure mechanism caused by metal movement under current stress.

TDDDB

Time-dependent dielectric breakdown; reliability mechanism affecting insulating layers.

HCI

Hot carrier injection; reliability mechanism that can degrade transistor performance.

NBTI

Negative-bias temperature instability; reliability mechanism affecting transistor threshold behavior.

Packaging

wire bond

Electrical connection made by bonding fine wires between die and package leads or substrate.

flip chip

Package approach where die are connected face-down through bumps.

underfill

Material placed under flip-chip die to improve mechanical reliability.

bump

Conductive connection point used in advanced packages or wafer-level packaging.

TSV

Through-silicon via used to connect vertically through silicon in advanced packages.

substrate

Package platform that supports the die and routes signals to the board.

Quality and customer**MRB**

Material review board that decides disposition for nonconforming or suspect material.

deviation

Approved exception from a normal requirement, route, specification, or process condition.

rework

Additional approved processing intended to bring material back into requirement.

scrap

Disposition for material that cannot be used or recovered acceptably.

FA

Failure analysis; structured investigation of why a device, die, package, or system failed.

decapsulation

Removal of package material to inspect die, bonds, or physical evidence during FA.

Expressions for this course

Complete the frames with the facts of your case. A frame helps organize meaning; it should not replace an actual explanation.

Make a complex point clear

In this context, ... means

The difference is that

How would you explain that distinction to a colleague?

Start with the reader's question. Explain one unfamiliar term with familiar words, then give a relevant example or contrast. Check understanding without asking only 'Do you understand?'.

Ask a precise question

When you say ..., do you mean ...?

Could you clarify which ...?

So, my understanding is Is that right?

Name the exact word, fact, or requirement you need clarified. Ask one focused question at a time. Repeat your understanding and give the other person a chance to correct it.

Match certainty to evidence

The available evidence shows

This may indicate ..., but

We have not yet established whether

Say what the evidence shows, identify what is still unknown, and limit the conclusion to that evidence. Words such as 'may', 'suggests', and 'in this sample' are useful when the uncertainty is real. Do not soften an urgent, verified safety concern.

Give a useful status update

We have completed ...; ... remains open.

As of ..., the status is

The next update will cover

Lead with the current state, then the important change or open item. Use completed-action language only for completed work. Add a next update point when it is known; do not invent a date to make the message sound complete.

Ask for an actionable next step

Could you provide ... so that ...?

Please confirm ... by [agreed time].

If that timing is not possible, please let me know

State the action and the information needed. Explain why it matters and specify a deadline only when one is given or agreed. 'Could you' is polite; repeated apologies can obscure a legitimate request. Urgent situations may require a direct request.

Explain a fair comparison

Both options ..., but

Option A ..., whereas option B

The comparison depends on

Use the same time period, scope, and measurement basis for both options. Explain a difference with 'whereas' or 'while'. State what improves and what is given up; do not hide the weaker side of a recommendation.

Offer a workable tradeoff

We can ..., provided that

Would you prefer ... or ...?

I can take that proposal for review; I cannot yet confirm

Acknowledge the goal and explain the available choices. Connect an offer to its actual conditions. A useful response makes room for discussion without promising approval or resources you do not control.

Model responses in context

Read the situation first. Cover the model, say your own response, and compare the two for meaning and tone.

01 | Wafer Fabrication Flow and Process Integration

A new colleague confuses wafer fabrication with packaging. A process overview needs to explain where the current work takes place.

Wafer fabrication forms devices on the wafer; packaging happens later in the product flow. Our update concerns the fabrication stage. I'll show the relevant process steps before discussing the issue.

Try it again: Your listener is new to this field. Replace two specialist expressions with clear explanations without changing the meaning.

02 | Lithography, Reticles, and Critical Dimensions

A lithography review reports a critical-dimension shift but does not identify the layer or measurement conditions.

Which layer and measurement conditions does the shift refer to? Please include the target, observed result, and reference recipe so we can discuss the same comparison.

Try it again: The other person uses an ambiguous word such as 'ready', 'approved', or 'urgent'. Clarify it before continuing.

03 | Deposition, Etch, CMP, and Process Windows

A process change improves one measured outcome but has not been checked across the proposed operating range.

The result is encouraging at the tested settings. We have not established the wider process window. I'll distinguish the observed improvement from the conditions that still need evaluation.

Try it again: Someone asks for a yes-or-no answer when the evidence supports only a qualified answer. Be concise while preserving the uncertainty.

04 | Metrology, SPC, and Yield Learning

A pilot lot has 92% yield compared with 90% in a reference lot. The sample sizes and product mix differ.

The pilot's reported yield is two percentage points higher. The lots differ in size and mix, so the comparison needs qualification. I'll show those details alongside the headline result.

Try it again: The listener asks you to reduce the update to two sentences. Preserve the most important completed and pending items.

05 | Defect Density and Cleanroom Contamination

A contamination review needs the timing and location of observed particles. The current note says only "cleanroom problem."

Could you provide the observation time, location, and relevant records? A precise description will help the team investigate without assuming a contamination source before reviewing the evidence.

Try it again: The other person cannot meet the requested timing. Clarify an alternative without inventing authority to change a formal deadline.

06 | Equipment Uptime, Recipes, and Tool Matching

Two tools run the same nominal recipe but show different measured results. A team member assumes their performance is interchangeable.

The recipe name is the same, but the observed results differ. Let's compare the controlled settings and qualification evidence before describing the tools as matched for this process.

Try it again: Your listener focuses only on the lowest price or fastest date. Explain one other relevant comparison criterion.

07 | Packaging, Test, and Reliability Qualification

A package passes one reliability test while two planned tests remain open. A customer slide calls qualification complete.

One test is complete; two are still open. I'll report the results by test and avoid describing the overall qualification as complete until the responsible team confirms the required evidence.

Try it again: Someone asks for a yes-or-no answer when the evidence supports only a qualified answer. Be concise while preserving the uncertainty.

08 | Foundry, Tape-Out, PDK, and Capacity Communication

A customer requests an earlier tape-out slot. Design checks are still open, and foundry capacity is limited.

We can review the earlier slot, but the open design checks and capacity need confirmation. Could we agree on a decision point that reflects both readiness and the foundry's available schedule?

Try it again: The preferred option is unavailable. Offer an alternative and clearly state any approval still needed.

Use the language responsibly

These cases practice communication. Actual equipment, construction, and safety work follows approved procedures and authorized specialist decisions.

Meaning before performance

Use specialist terms only when they help the listener. Explain unfamiliar words, preserve uncertainty, and ask when an instruction or responsibility is unclear. The model responses illustrate language; they do not authorize professional actions.

Sources and further reading

The cases, workshops, and explanations are original teaching material. These references provide language frameworks and selected professional context. References were checked in September 2026; consult current local guidance for actual work.

Digital.gov: plain-language guidance

<https://digital.gov/guides/plain-language>

Council of Europe: CEFR descriptors

<https://www.coe.int/en/web/common-european-framework-reference-languages/cefr-descriptors>

Your next step

Choose one expression you can use in a genuine work situation. Rehearse it with fictional details, then adapt the wording to your role and audience.